

Part 2: The Autoregressive Models

Text references: §12.4 – 12.6 in Ruppert and Matteson

A simple time series model is the *autoregressive (AR) process*. In this model, X_t is modelled as a linear combination of the p most recent values of the process $(X_{t-1}, X_{t-2}, \dots, X_{t-p})$ plus some random error.

The value of p is called the *order* of the model. The use of p to denote this is quite standard, and can help to avoid confusion with other, related quantities which will appear later.

The term *autoregressive* arises from the fact that one can think of fitting a regression model to estimate the parameters.

The AR(1) Model

We begin by discussing the simple AR(1) model.

Assume that $\{\epsilon_t\}$ is WN(0, σ_ϵ^2) with $\sigma_\epsilon^2 > 0$. Then $\{X_t\}$ is an AR(1) process if for some parameters μ and ϕ it holds that

$$X_t - \mu = \phi(X_{t-1} - \mu) + \epsilon_t$$

We can see that the value of the process at time t is constructed as the sum of

1. μ , the mean of the process,
2. $\phi(X_{t-1} - \mu)$, the contribution from the previous time step, and
3. ϵ_t , an error term.

The term ϵ_t is meant to model the contribution from the information that is only available at time t , and not previously.

1 It is useful to note that the AR(1) model can be rewritten as

2
$$X_t = \beta_0 + \beta_1 X_{t-1} + \epsilon_t$$

3 where $\beta_0 = (1 - \phi)\mu$ and $\beta_1 = \phi$.

4 This expression makes it clear how this model can be thought of as a
5 regression model.

- 1 **Exercise:** Imagine that we can extend the process back an infinite
 2 amount in time. Show that if $|\phi| < 1$, then

$$X_t = \mu + \sum_{h=0}^{\infty} \phi^h \epsilon_{t-h}$$

$$\begin{aligned}
 X_t &= \beta_0 + \beta_1 X_{t-1} + \epsilon_t \\
 &= (1-\phi)\mu + \phi X_{t-1} + \epsilon_t \\
 &= (1-\phi)\mu + \phi((1-\phi)\mu + \phi X_{t-2} + \epsilon_{t-1}) + \epsilon_t \\
 &= (1-\phi)\mu + \phi(1-\phi)\mu + \phi^2 X_{t-2} + \phi\epsilon_{t-1} + \epsilon_t \\
 &= \sum_{k=0}^{\infty} \phi^k (1-\phi)\mu + \sum_{k=0}^{\infty} \phi^k \epsilon_{t-k} \\
 &= \mu + \sum_{k=0}^{\infty} \phi^k \epsilon_{t-k} \quad \text{since } |\phi| < 1
 \end{aligned}$$

- 18 Because of the above result, the AR(1) model can be called a *moving*
 19 *average* model of infinite order.

1 The case where $\phi = 1$ is the random walk model.

2 **Exercise:** Show that the AR(1) model is (weakly) stationary if and
3 only if $|\phi| < 1$.

4 Assume $|\phi| < 1$.

5 $E(X_t) = \mu$ for all t since

$$6 \quad E\left(\mu + \sum_{k=0}^{\infty} \phi^k \epsilon_{t-k}\right) = \mu$$

$$7 \quad \text{Cov}(X_t, X_{t+k}) = E(X_t X_{t+k}) - E(X_t) E(X_{t+k})$$

$$8 \quad E(X_t X_{t+k}) = E\left[\left(\mu + \sum_{h=0}^{\infty} \phi^h \epsilon_{t-h}\right)\left(\mu + \sum_{h'=0}^{\infty} \phi^{h'} \epsilon_{t+k-h'}\right)\right]$$

$$9 \quad = \mu^2 + E\left[\left(\sum_{h=0}^{\infty} \phi^h \epsilon_{t-h}\right)\left(\sum_{h'=0}^{\infty} \phi^{h'} \epsilon_{t+k-h'}\right)\right]$$

$$10 \quad = \mu^2 + E\left[\left(\sum_{h=0}^{\infty} \phi^h \epsilon_{t-h}\right)\left(\sum_{h''=-k}^{\infty} \phi^{h''+k} \epsilon_{t-h''}\right)\right] \quad h'' = h' - k$$

$$11 \quad = \mu^2 + E\left[\sum_{h=0}^{\infty} \phi^{2h+k} \epsilon_{t-h}^2\right] \quad E(\epsilon_i \epsilon_j) = 0 \quad i \neq j$$

$$12 \quad = \mu^2 + \sum_{h=0}^{\infty} \phi^{2h+k} \sigma_{\epsilon}^2$$

$$13 \quad = \mu^2 + \phi^k \sigma_{\epsilon}^2 \sum_{h=0}^{\infty} \phi^{2h}$$

$$= \mu^2 + \sigma_\epsilon^2 \phi^k \left(\frac{1}{1-\phi^2} \right) \quad \text{since } |\phi| < 1$$

$$\text{So, } \text{Cov}(X_t, X_{t+k}) = \sigma_\epsilon^2 \phi^k \left(\frac{1}{1-\phi^2} \right)$$

Does not depend on t , so conclude that $\{X_t\}$ is stationary

If $|\phi| \geq 1$, then $V(X_t)$ grows with t (think about the random walk, $\phi=1$)

So, it is not stationary

1 More Properties of AR(1) Process

2 If the AR(1) process $\{X_t\}$ is stationary, then

3 1. $E(X_t) = \mu$ for all t

4 2. $\gamma(0) = V(X_t) = \sigma_\epsilon^2 / (1 - \phi^2)$

5 3. $\gamma(h) = \sigma_\epsilon^2 \phi^{|h|} / (1 - \phi^2)$

6 4. $\rho(h) = \phi^{|h|}$

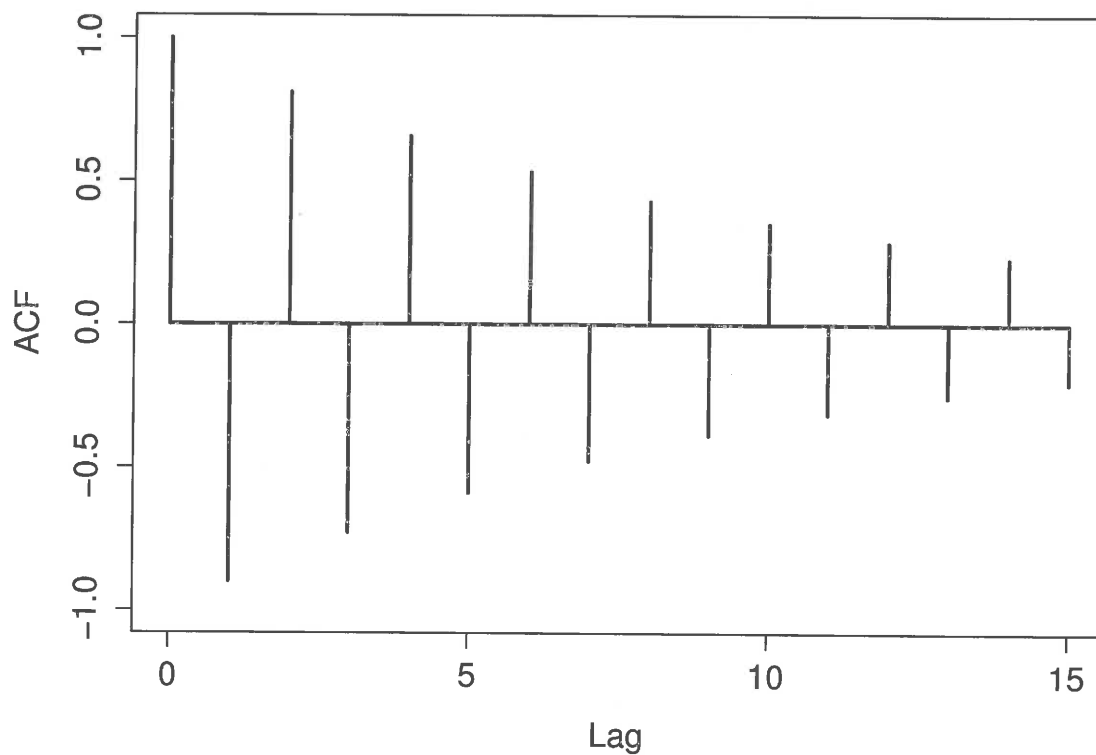


Figure 1: The theoretical ACF for the AR(1) process with $\phi = -0.9$.

1 **Exercise:** What is the (theoretical) PACF of the AR(1) process?

2 X_t is only a function of X_{t-1} , and
 3 not $X_{t-2}, X_{t-3}, \dots$ But X_t and X_{t-2}
 4 are still correlated (see previous plot).
 5 There is no relationship between X_t and
 6 X_{t-2} , conditional on X_{t-1} .

7
 8 So, the PACF equals 0 for all lags
 9 greater than one

10 The PACF at lag 1 is ϕ (same as ACF)
 11

12 This is useful for determining if AR(1)
 13 is appropriate
 14
 15
 16
 17

1 The Initial Value

2 When we wrote above that the AR(1) model is stationary when $|\phi| <$
3 1, this was meant to imply the case where we imagine the process
4 going back an infinite amount in time.

5 If we take X_0 to be the initial value of the series, then the stationarity
6 of the process $(X_0, X_1, \dots)$ depends on the distribution of X_0 .

7 In particular, if X_0 has mean μ and variance $\sigma_\epsilon^2/(1 - \phi^2)$, then the
8 process is stationary. (Assuming, again, that $|\phi| < 1$.)

9 Otherwise, the process will only *converge to the stationary distribution*.

10 **Exercise:** Explain why the series will typically appear stationary af-
11 ter only a few time steps.

12 In practice, X_t depends on X_{t-1} but also
13 a large contribution from ϵ_{t+1} , so after
14 a few time steps, the dependence on X_0
15 is mostly lost. So, the dist. of X_0
16 has little effect on the dist. of X_t
17 for t , even at small t . (Recall that the
correlation at lag h is ϕ^h)

2 Fitting The AR(1) Model

3 Fitting an AR(1) model in R can be accomplished using the function
4 `arima()` with argument `order=c(1, 0, 0)`.

p d q

5 (We will use this function repeatedly for fitting more general models;
6 the value of `order` will then change.)

7 There are two options for estimating the parameters: maximum like-
8 lihood and conditional least squares. These are described in Section
9 12.5.2 in Ruppert. See the argument `method` to `arima()` in order to
10 adjust this.

11 The default in R is to use conditional least squares to find the starting
12 value, but then complete the search using maximum likelihood.

1 Residuals

2 If we rearrange the expression for the AR(1) model, we see that

3
$$\epsilon_t = (Y_t - \mu) - \phi(Y_{t-1} - \mu)$$

4 and hence it is natural to define the *residuals* as

5
$$\hat{\epsilon}_t = (Y_t - \hat{\mu}) - \hat{\phi}(Y_{t-1} - \hat{\mu})$$

6 for $t \geq 2$.

7 If the model is correct, then the residuals should be approximately
8 uncorrelated.

9 Hence, the Ljung-Box test is a useful tool for testing the validity of
10 this assumption.

11 **BUT:** One needs to adjust the test to account for the fact that addi-
12 tional parameters have been estimated; this is done in R using the ar-
13 gument `fitdf` to `Box.test()`. In particular, when fitting an AR(p)
14 model, one should set `fitdf = p`.

1 It is also useful to inspect the time series of the residuals, to look for
2 patterns or trends.

3 Finally, the normal probability plot of the residuals can be used to
4 get an idea of the distribution of the errors.

5 **Exercise:** Why might it be important to test the validity of the nor-
6 mality assumption?

7 If using MLE and assuming normal ϵ_t . (The
8 default assumption using `arima()`.)

9
10 If using AIC to compare models, and
11 assuming ϵ_t are normal, then recall
12 that the validity of AIC rests on
13 this assumption being correct

14
15

$$AIC = -2 \log l_{\text{the}} + 2p$$

R Exercise: Obtain the daily market data for IBM for January 1, 2017 to June 30, 2018. We will consider here the log daily trading volume over this time period.

1. From looking at the time series, do you believe that this series is stationary? *Questionable, but seems like a reasonable approx.*
 2. Inspect the PACF of this series. Does an AR(1) series seem a reasonable choice for a model? *Strong correlation at lag 1, but no evidence of correlation beyond. So,*
 3. Fit an AR(1) model to these data. What does the estimate of the parameter ϕ tell you about the stationarity of this series? *AR(1) is reasonable*
 4. Inspect the residuals from the fit and comment on them. Look at both the normal probability plot and the ACF. *Since $|\phi| = 0.5812$ is not close to 1, does not suggest not stationary. Normal prob. plot suggests upper tail heavier than normal. ACF of residuals does not show remaining correlation.*
 5. Use the Ljung-Box Test on the residuals, and interpret the results. *Remember to use the argument `fitdf`. If test out to lag of 20, p-value is 0.7148, i.e. no evidence of remaining correlation.*
- In R output:

ar1	"intercept"	
0.5812	15.2373	$\hat{\sigma}_e^2 = 0.1008$
$\hat{\phi}$	$\hat{\mu}$	
	$\hat{\sigma}_e^2$	

The AR(p) Model

R Exercise: Use the following sequence of R commands to obtain five years of data on the 3-Year Treasury Constant Maturity Rate (daily, not seasonally adjusted). Then, the data are transformed into the first differences. Does the AR(1) model seem like a good fit in this case?

```
> DGS3 = getSymbols("DGS3", src="FRED",  
                    auto.assign=F)  
> DGS3 = last(DGS3, '5 years')  
> DGS3diff = diff(DGS3, 1, 1)  
> DGS3diff = DGS3diff[!is.na(DGS3diff)]
```

If anything, need to go out to a further
lag (of 5) in order to get a
good fit.

1 The PACF of the above time series suggests that there could be de-
2 pendence on more than just the previous day's value of the series.

3 This is often the case, and motivates the consideration of AR models
4 of order $p > 1$.

5 The general form for the $AR(p)$ model is as follows:

$$6 \quad X_t - \mu = \sum_{i=1}^p \phi_i (X_{t-i} - \mu) + \epsilon_t$$

7 or, equivalently,

$$8 \quad X_t = \beta_0 + \sum_{i=1}^p \phi_{t-i} X_{t-i} + \epsilon_t$$

9 where $\beta_0 = (1 - \sum_i \phi_i)\mu$. Here, $\{\epsilon_t\}$ is assumed to be a white noise
10 process.

11 This second form makes it look like a familiar multiple regression
12 model.

13 The choice of p will introduce many of the same issues of overfitting
14 that we saw in the case of regression; indeed, tools such as AIC will
15 be used to make this selection.

1 The ACF and PACF of the AR(p) Process

2 The form for the ACF in this case is complicated, but R function
3 `ARMAacf()` allows one to explore how the ACF or PACF varies with
4 the values of ϕ_i .

5 For example, the commands

ϕ_1 ϕ_2

```
6 > plot(0:10, ARMAacf(ar=c(0.5, -0.2), lag.max=10),  
7       type="b", pch=16)  
8 > plot(1:10, ARMAacf(ar=c(0.5, -0.2), lag.max=10,  
9       pacf=TRUE), type="b", pch=16)
```

10 show the ACF and PACF for the case of an AR(2) model with $\phi_1 = 0.5$
11 and $\phi_2 = -0.2$.

12 **An important property:** The PACF of an AR(p) model at lag h is
13 equal to zero for $h > p$. As discussed above, this is a useful tool for
14 model selection.

1 Fitting the $AR(p)$ Model

2 The fitting of the $AR(p)$ model proceeds much as for the $AR(1)$ model,
3 but now with `order = c(p, 0, 0)`.

4 The residuals should be inspected in an identical manner as was
5 done in the $AR(1)$ case.

6 As was mentioned above, AIC is often used to choose p . The R func-
7 tion `auto.arima()`, part of package `forecast`, enables the search
8 over different values of p , finding the optimal model using AIC.

1 A few comments on the use of this function:

2 1. The argument `max.p` sets the largest value of p to be considered.

3 2. To fit a pure $AR(p)$ model, one must set `max.q=0`.

4 3. To use AIC, set `ic="aic"`. The default is `ic="aicc"`, the "cor-
5 rected" version of AIC.

6 4. The data passed into this function must be a vector, not an `xts`
7 object.

8 5. By default, when the time series is sufficiently long, an approxi-
9 mation algorithm is utilized and the model found to be optimal
10 may not be best when `approximation = F`. See, for example,
11 Question 2 on page 356 of Ruppert.

12 **R Exercise:** Reconsider the log daily trading volume for IBM. What
13 order AR model is chosen when using AICc as the criterion?

14 Set `max.d=0` to avoid any differencing

15 Set `max.q=0` to exclude MA component

16
17 With or without approx. algorithm, conclude
18 that $AR(1)$ is best choice among $AR(p)$
19 models.

Stationarity of the $AR(p)$ Process

First, recall that a *root* of a function of x , say $f(x)$, is any value for x such that $f(x) = 0$.

It can be shown (it's technical) that stationarity of the $AR(p)$ process is equivalent to the condition that every root of the polynomial

$$f(x) = 1 - \sum_{i=1}^p \phi_i x^i$$

has absolute value (modulus) greater than one.

So, when $p = 1$, we note that $1 - \phi x$ has a single root of $1/\phi$, and the condition $|1/\phi| > 1$ is equivalent to $|\phi| < 1$.

1 **Example (Example 12.12 in Ruppert):** In this example, a model was
2 fit with $\hat{\phi}_1 = 1.2074$ and $\hat{\phi}_2 = -0.2237$. If we set up the equation

3
$$1 - \hat{\phi}_1 x - \hat{\phi}_2 x^2 = 0$$

4 we can find the roots using `polyroot()`:

5 `> polyroot(c(1, -1.2074, 0.2237))`

6 `[1] 1.021584+0i 4.375823-0i`

7 These are both larger than one in absolute value, but the first is quite
8 close. Ruppert states, “since the roots are estimated with error, there
9 is reason to suspect that this series may be nonstationary.”

Unit Root Tests

The case where one or more of the above-mentioned roots are equal to one is the borderline between stationarity and nonstationarity. This is called the presence of *unit roots*.

There exist hypothesis tests for stationarity, called *unit root tests*.

The simplest version, the *Dickey-Fuller Test*, is based on a simple AR(1) model:

$$Y_t = \phi Y_{t-1} + \epsilon_t$$

and the test is $H_0: \phi = 1$ versus $H_1: \phi < 1$.

Exercise: Describe the behavior of the series in the case that $\phi = 1$, i.e., under the null hypothesis.

When $\phi=1$, $Y_t = Y_{t-1} + \epsilon_t$, a random walk
(not stationary)

Let $\Delta Y_t = Y_t - Y_{t-1}$

If $\phi=1$, $\Delta Y_t = \epsilon_t$, i.e. $\Delta Y_t = \epsilon_t$ is
white noise (stationary)

- 1 Keep in mind that the basic Dickey-Fuller test described above tests
2 a random walk versus a stationary AR(1) model.

3 Extensions of this test allow for the possibility of a *deterministic trend*.

4 A series $\{Y_t\}$ is said to be *trend stationary* if

5
$$Y_t = f(t) + \epsilon_t$$

6 where $f(t)$ is a deterministic function and $\{\epsilon_t\}$ is stationary.

7 For instance, one version of the Dickey-Fuller Test uses the model

8
$$Y_t = \underbrace{\beta_0 + \beta_1 t}_{f(t)} + \phi Y_{t-1} + \epsilon_t$$

9 allowing for a linear deterministic trend.

1 Another important extension of this test is the widely used *augmented*
2 *Dickey-Fuller (ADF) test*.

3 To derive this, it is standard to first rewrite the model underlying the
4 original Dickey-Fuller as

$$\Delta Y_t = \pi Y_{t-1} + \epsilon_t$$

$$\begin{aligned} Y_t &= \phi Y_{t-1} + \epsilon_t \\ Y_t - Y_{t-1} &= (\phi - 1) Y_{t-1} + \epsilon_t \\ \Delta Y_t &= \pi Y_{t-1} + \epsilon_t \end{aligned}$$

5
6 where $\Delta Y_t = Y_t - Y_{t-1}$ and $\pi = \phi - 1$. The null hypothesis then
7 becomes $H_0: \pi = 0$.

8 Now, expand the model to allow for an intercept and a linear trend,
9 plus dependence on time points up to lag $k + 1$:

$$\Delta Y_t = \beta_0 + \beta_1 t + \pi Y_{t-1} + \sum_{i=1}^k \gamma_i \Delta Y_{t-i} + \epsilon_t$$

$$Y_{t-i} - Y_{t-i-1}$$

10
11 The test again uses $H_0: \pi = 0$. Under the null, the series $\{\Delta Y_t\}$ is
12 $AR(k)$ (after the deterministic trend is removed).

13 When $\pi < 0$, the series $\{Y_t\}$ is trend stationary $AR(k)$.

14 When $\pi > 0$, the series $\{Y_t\}$ is said to be *explosive*, in that the series
15 goes off to $-\infty$ or ∞ .¹

¹There is a difference between a series being nonstationary and being explosive. A random walk is nonstationary, but not explosive, as a random walk will return to (cross) the starting value infinitely often with probability one. An explosive process will cross the starting value a finite number of times.

1 The function `adf.test()`, part of the package `tseries`, imple-
2 ments the ADF test.

3 The user can select from between two alternative hypotheses: Either
4 that the series is stationary ($\pi < 0$) or that the series is “explosive”
5 ($\pi > 0$). The alternative is chosen via the argument `alternative`,
6 with value of either `stationary` or `explosive`.

7 The value of k can be specified using the argument `k`. If `k=0` is used,
8 then the standard Dickey-Fuller test results. The default value of k
9 is approximately the cube root of the length of the series, but one
10 should make sure that a reasonable selection is made.

11 **NOTE:** The object passed on to this function must be a vector, i.e. wrap
12 any `xts` object with `as.numeric()`.

- 1 The results of this test applied to our IBM and 3-Year Treasury data
2 can be found as follows:

```
3 > adf.test(as.numeric(log(Vo(IBM))), alternative="s")  
4 > adf.test(as.numeric(DGS3diff), alternative="s")
```

5 **Exercise:** What are the conclusions of the above tests?

6 For log volume, $p\text{-value} < 0.01$, so have
7 evidence the series is stationary

8 Likewise for DGS3diff
9
10
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Part 3: MA, ARMA, ARIMA

Text references: §12.7 – 12.9, 12.11 in Ruppert and Matteson

Recall from Part 2 that we found that an AR(1) model can be written in the form

$$X_t = \mu + \sum_{h=0}^{\infty} \phi^h \epsilon_{t-h}$$

and is hence referred to as a *moving average model of infinite order*.

A general *moving average of order q* ($MA(q)$) is written as

$$X_t = \mu + \epsilon_t + \sum_{j=1}^q \theta_j \epsilon_{t-j}$$

where $\{\epsilon_t\}$ is $WN(0, \sigma_\epsilon^2)$.

Exercise: What is the motivation for the name “moving average?”

Consider ϵ_t to be the “new information” available at time t . X_t is a linear combination of the new info. plus the preceding q bits of info.

1 Properties of MA(q) Models

2 Assume that $\{X_t\}$ is a MA(q) process. Then:

3 1. The process is (weakly) stationary.

4 2. $E(X_t) = \mu$

5 3. $V(X_t) = \sigma_\epsilon^2 (1 + \theta_1^2 + \theta_2^2 + \dots + \theta_q^2)$

6 4. $\gamma_X(h) = \rho_X(h) = 0$ for $|h| > q$.

7 5. In the case of $q = 1$, we can show that

8 (a) $\gamma_X(1) = \theta_1 \sigma_\epsilon^2$

9 (b) $\rho_X(1) = \theta_1 / (1 + \theta_1^2)$

10 **Exercise:** Give a practical interpretation of Property 4 above. How
11 could the sample ACF aid in the selection of q ?

12 Plot the sample ACF of the series.
13 See if there is some m such that
14 there is no remaining correlation
15 beyond lag m . Could try $q=m$.

16

17

18

19

1 **Exercise¹:** Assume that $\{\epsilon_t\}$ is $WN(0, \sigma_\epsilon^2)$. Derive the theoretical ACF
2 for the time series model

3
$$Y_t = \epsilon_t - \theta_1 \epsilon_{t-1}.$$

4 Then, derive the theoretical ACF for the model

5
$$X_t = \epsilon_t - \frac{1}{\theta_1} \epsilon_{t-1}.$$

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¹This is a technical comment based on Example 3.1 in Chan.

1 The previous example illustrates that the different forms can yield
2 processes with the same random behavior.

3 To make a choice between these forms, consider the following.

4 We can rewrite

$$5 \qquad \epsilon_t = Y_t + \theta_1 \epsilon_{t-1} \qquad (1)$$

$$6 \qquad \qquad \qquad = Y_t + \theta_1 (Y_{t-1} + \theta_1 \epsilon_{t-2}) \qquad (2)$$

$$7 \qquad \qquad \qquad = Y_t + \theta_1 Y_{t-1} + \theta_1^2 Y_{t-2} + \dots \qquad (3)$$

8 and we can likewise write

$$9 \qquad \qquad \qquad \epsilon_t = X_t + \frac{1}{\theta_1} \epsilon_{t-1}$$
$$10 \qquad \qquad \qquad = X_t + \frac{1}{\theta_1} X_{t-1} + \frac{1}{\theta_1^2} X_{t-2} + \dots$$

11 What we note here is that if $|\theta_1| < 1$, then the coefficients in the ex-
12 pansion of ϵ_t shown in line (3) will shrink to zero as we move further
13 back in time. This is a desirable property, and in such a case we say
14 that the process is *invertible*.

15 Thus, it is standard to write the MA(1) in the form shown in line (1),
16 and assume that $|\theta_1| < 1$.

1 Example of Fitting an MA(q) Process

- 2 Here we simulated a time series of length 250 from a MA(2) process
3 with $\theta_1 = 0.3$, $\theta_2 = -0.2$, and $\sigma_\epsilon^2 = 1$. The series is shown as Figure 1,
4 and its sample ACF is shown as Figure 2.

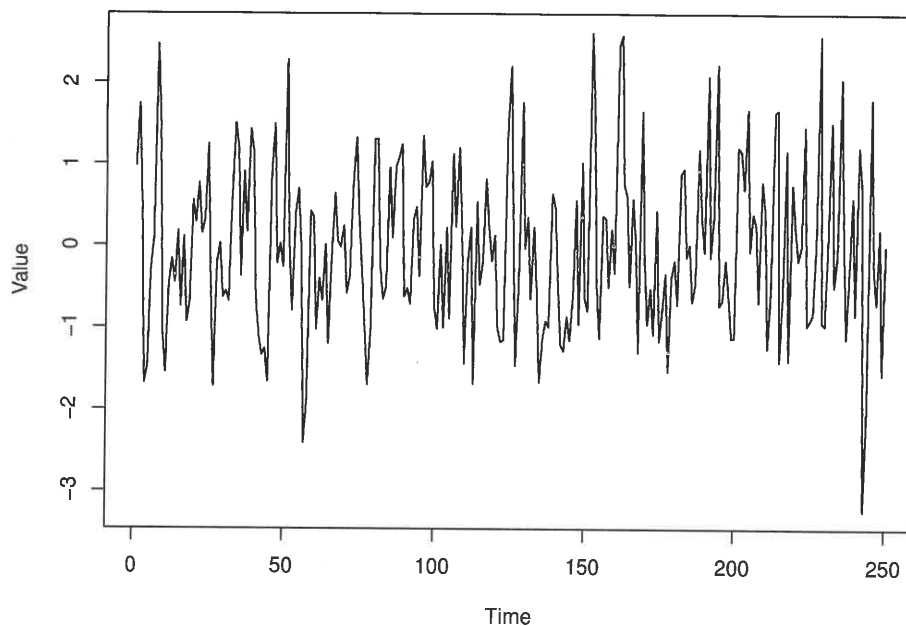


Figure 1: Simulated time series from a MA(2) model.

- 5 The results of the ADF test are shown below.

```
6         Augmented Dickey-Fuller Test
```

```
7
```

```
8 data:  as.numeric(simdat)
```

```
9 Dickey-Fuller = -6.1653, Lag order = 6, p-value = 0.01
```

```
10 alternative hypothesis: stationary
```

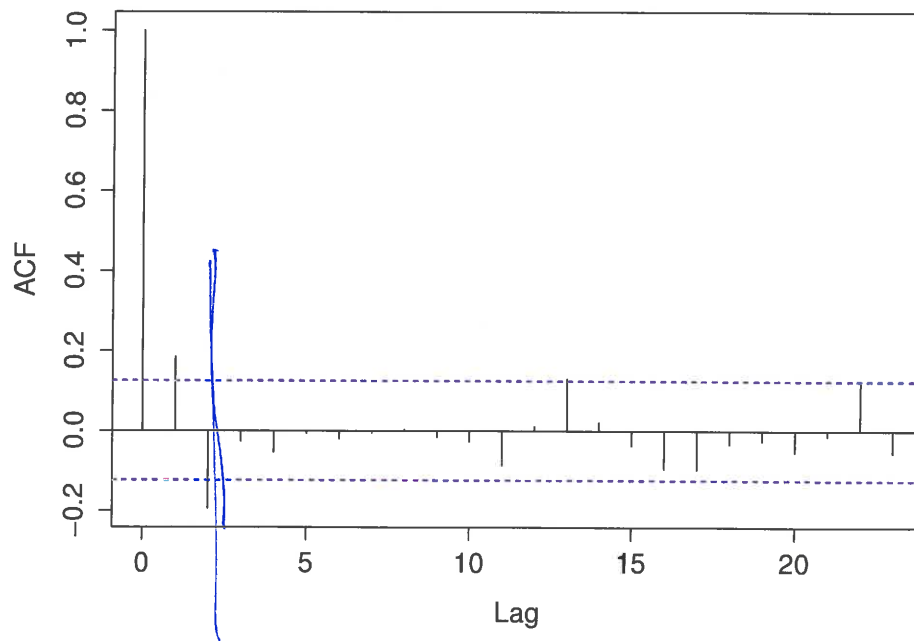


Figure 2: The sample ACF for the simulated time series.

- 1 **Exercise:** Explain why the MA(2) model is a reasonable choice in this
2 situation.

3 Strong correlation at lags 1 and 2, but
4 none beyond. Suggests MA(2)

- 1 The model is fit in R using the `arima()` function. Note how the
 2 argument `order` is set: the third entry in the list is the value of q :

```
3 > holdfit = arima(simdat, order=c(0,0,2))
```

$p d q$

- 4 The results from the fit are shown below:

5 Call:

```
6 arima(x = simdat, order = c(0, 0, 2))
```

7

8 Coefficients:

	$\hat{\theta}_1$	ma1	$\hat{\theta}_2$	ma2	intercept	$\hat{\mu}$
	0.2648		-0.2468		-0.0112	
s.e.	0.0624		0.0651		0.0616	

12

13 sigma² estimated as $\hat{\sigma}_e^2$ 0.9134: log likelihood = -343.54,
 14 aic = 695.08

- 15 **Exercise:** Construct a 95% confidence interval for θ_1 .

$$0.2648 \pm (2)(0.0624)$$

1 The result from the Ljung-Box Test applied to the residuals of the
2 fit are shown below. Note that `fitdf = 2` to account for the two
3 additional parameters that have been estimated.

```
4 > Box.test(holdfit$residuals, fitdf=2, lag=20,  
5           type="Ljung-Box")
```

6
7 Box-Ljung test

8
9 data: holdfit\$residuals

10 X-squared = 11.0549, df = 18, p-value = 0.892

11 **Exercise:** Explain the conclusion of the test above. What other steps
12 should be taken with the residuals?

13 Box test suggests no evidence of further
14 correlation

15 Look at normal prob. plot, also ACF,
16 of residuals
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- Figure 3 compares the sample ACF with the ACF that results from the model fit. In this case, we can also plot the true ACF; that is shown on the plot as the solid circles.

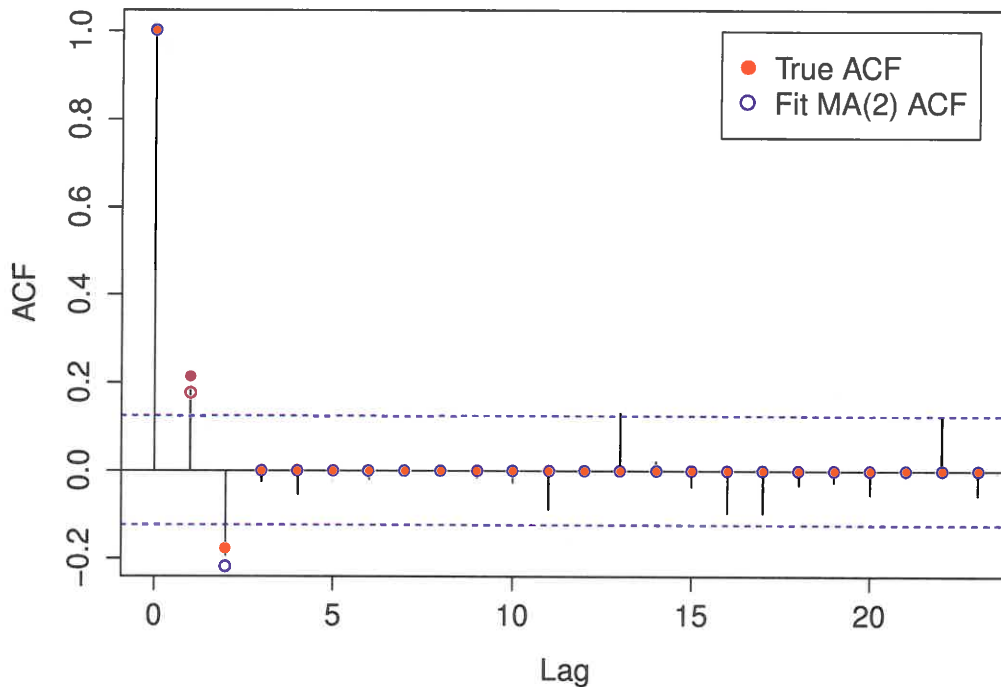


Figure 3: The sample ACF for the simulated time series, with the true ACF, and the ACF as fit from the MA(2) model.

- Here is the R code used to create this plot.

```

> acf(simdat, cex.axis=1.3, cex.lab=1.3, main="")
> trueacf = ARMAacf(ma = c(0.3, -0.2), lag.max=30)
> points(0:30, trueacf, pch=16, col = 2)
> fitacf = ARMAacf(ma = holdfit$coef[1:2], lag.max=30)
> points(0:30, fitacf, col = 4)
> legend(23, 1, legend=c("True ACF", "Fit MA(2) ACF"),
      pch=c(16, 1), col=c(2, 4), xjust=1, cex=1.3)

```